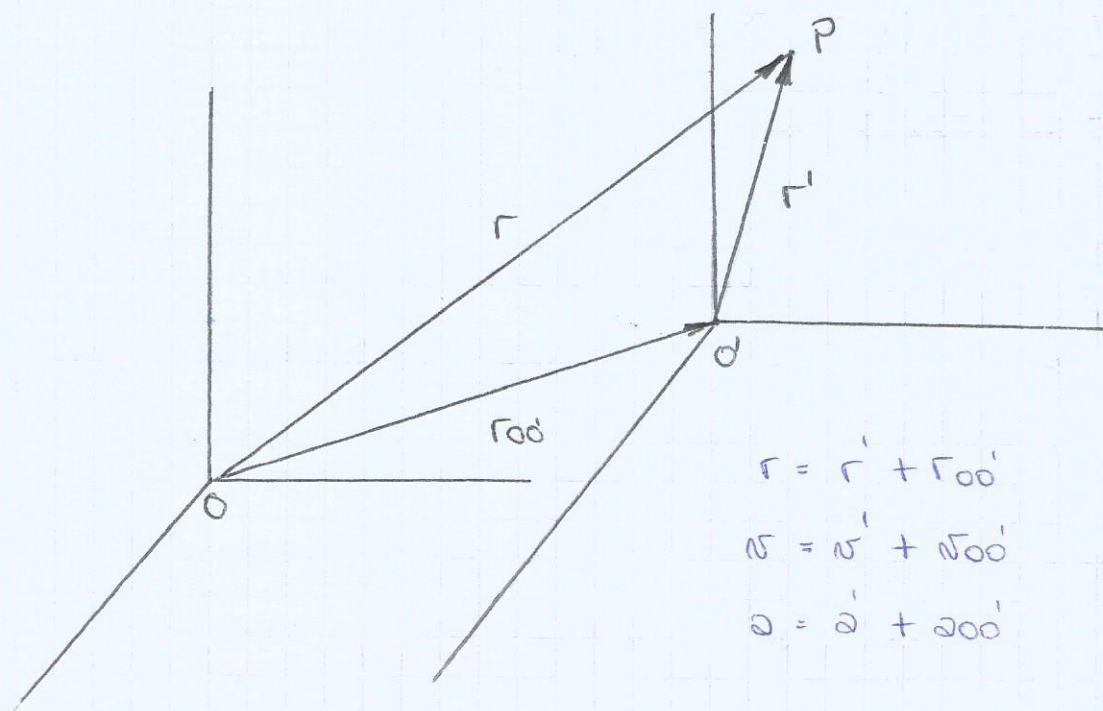


Unidad IX: Sistemas no inerciales



Si O es un sistema inercial

$$a = \frac{F}{m}, \text{ entonces } a' = \frac{F}{m} - a_{OO'}$$

- Si $a_{OO'} = 0 \Rightarrow O'$ es S.I
- Si $a_{OO'} \neq 0 \Rightarrow O'$ es S.N.I

Si O es S.I todo sistema que se mueve con MRU respecto de O es S.I

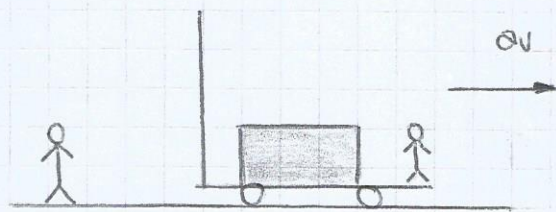
En un sistema no inercial: $ma' = F - \underbrace{ma_{OO'}}_{\text{Fuerza ficticia}}$

Fuerza ficticia

$$f^* = -ma_{OO'}$$

$$ma' = F + f^*$$

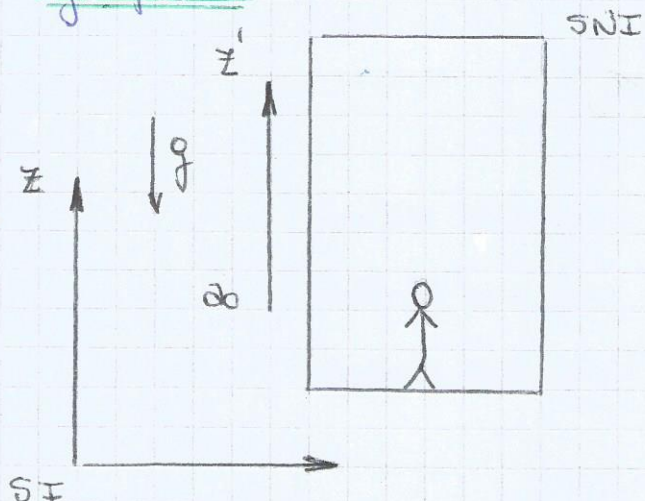
Ejemplo 1:



$$m a_B' = f^* = -m a_v$$

$$a_B' = -a_v$$

Ejemplo 2:



SN I: ascensor

SI: hombre en la Tierra

$$S.I) N - P = m a_z$$

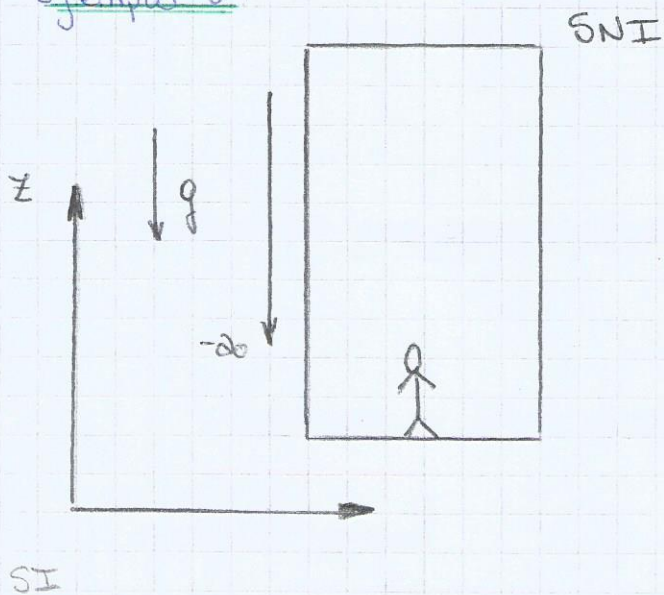
$$N - P = m a_0$$

$$S.N.I) \underbrace{N - P}_F - \underbrace{m a_0}_{f^*} = 0$$

Vínculo: $a_z = a_0$

$$z_h = z_a$$

Ejemplo 3:



SN I

$$S.I) N - P = -m a_0$$

$$S.N.I) N - P + \underbrace{m a_0}_{f^*} = 0$$

• Caso I: $a_A = -g \hat{z}$

$$S.N.I: N - P + mg = 0$$

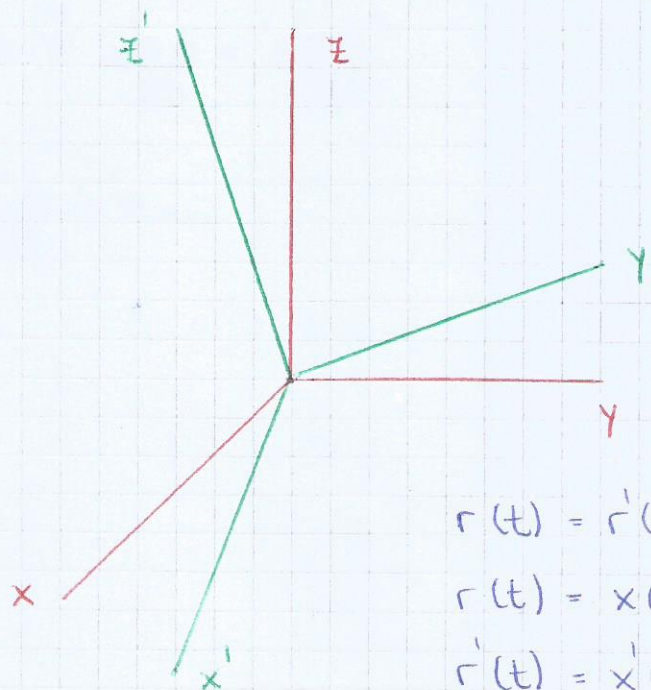
$$N = 0$$

• Caso II: $a_A = -2g \hat{z}$

$$S.N.I: N - P + 2mg = 0$$

$$-mg + 2mg = m a_H$$

Sistemas no inerciales rotantes:



/// Sistema inercial

/// Sistema no inercial

$$r(t) = r'(t)$$

$$r(t) = x(t) \hat{x} + y(t) \hat{y} + z(t) \hat{z}$$

$$r'(t) = x'(t) \hat{x}' + y'(t) \hat{y}' + z'(t) \hat{z}'$$

$$\frac{dr}{dt} = \frac{dr'}{dt} \quad v = \frac{dr}{dt} = \frac{dx}{dt} \hat{x} + \frac{dy}{dt} \hat{y} + \frac{dz}{dt} \hat{z}$$

$$\frac{dr'}{dt} = \frac{dx'}{dt} \hat{x}' + x' \frac{d\hat{x}'}{dt} + \frac{dy'}{dt} \hat{y}' + y' \frac{d\hat{y}'}{dt} + \frac{dz'}{dt} \hat{z}' + z' \frac{d\hat{z}'}{dt}$$

$$v' = \frac{dx'}{dt} \hat{x}' + \frac{dy'}{dt} \hat{y}' + \frac{dz'}{dt} \hat{z}'$$

$$\frac{d\hat{x}'}{dt} = \omega \times \hat{x}'$$

$$\frac{dx'}{dt} = \frac{dx'}{dt} \hat{x}' + x' \frac{d\hat{x}'}{dt}$$

$$\text{Con: } \frac{d\hat{x}'}{dt} = \omega \times \hat{x}'$$

$$= v' \hat{x}' + x' \vec{\omega} \times \hat{x}'$$

$$= v' \hat{x}' + \omega \times x'$$

$$= v' + \omega \times r'$$

$$\frac{dr}{dt} = \frac{dr'}{dt}$$

$$v = v' + \omega \times r'$$

$$\frac{dv}{dt} = \frac{dv'}{dt} + \frac{d}{dt}(\omega \times r')$$

$$a = \frac{dv}{dt}$$

$$\frac{dv'}{dt} = \frac{dv'_x}{dt} \hat{x}' + v'_x \frac{d\hat{x}'}{dt}$$

$$a'_x + v'_x \omega \times \hat{x}' = a'_x + \omega \times v'_x$$

$$\frac{dv'}{dt} = a' + \omega \times v'$$

$$\frac{d}{dt}(\omega \times r) = \dot{\omega} \times r + \omega \times \dot{r}$$

$$= \dot{\omega} \times r + \omega \times v$$

$$= \dot{\omega} \times r' + \omega \times (v' + \omega \times r')$$

$$= \dot{\omega} \times r' + \omega \times v' + \omega \times (\omega \times r') \quad r \perp \omega$$

$$= \dot{\omega} \times r' + \omega \times v' + \omega \times (\omega \times r_{\perp})$$

Considerando: $A \times (B \times C) = B(AC) - C(AB)$

$$\omega \times (\omega \times r_{\perp}) = \omega(\omega \cdot r_{\perp}) - r_{\perp}(\omega \cdot \omega)$$

$$\omega \times (\omega \times r_{\perp}) = -r_{\perp} \omega^2$$

$$\frac{dv}{dt} = a(t) = a' + \omega \times v' + \dot{\omega} \times r + \omega \times v' - r_{\perp} \omega^2$$

$$a(t) = a' + 2\omega \times v' + \dot{\omega} \times r - r_{\perp} \omega^2$$

$$a'(t) = a(t) + \underbrace{2v' \times \omega}_{\text{aceleración de Coriolis}} + \underbrace{r \times \dot{\omega}}_{\text{aceleración centrífuga}} + r_{\perp} \omega$$

aceleración
de Coriolis.

aceleración
centrífuga

$$m \ddot{a}'(t) = m \ddot{a}(t) + \underbrace{2m \dot{r}' \times \omega + m r' \times \dot{\omega} + m r' \perp \omega^2}_{\text{Fuerzas ficticias}}$$

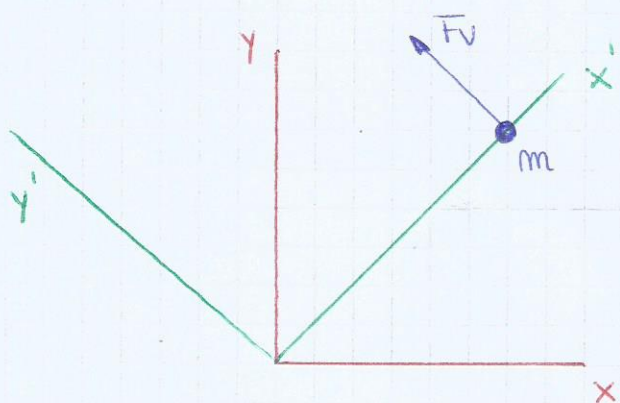
Fuerzas ficticias que el observador NI necesita incorporar para explicar el movimiento que ve.

$$f_c^* = 2m \dot{r}' \times \omega \quad (\text{Solo si } P \text{ se mueve respecto de } O')$$

$$f_r^* = m r' \times \dot{\omega}$$

$$f_{cf}^* = m r' \perp \omega^2 \quad (\text{Siempre para un sistema rotante}).$$

Ejemplo 1:



$$\omega = \text{cte}$$

/// Sistema inercial

/// Sistema no inercial

$$\otimes g$$

a) S.I fijo a la mesa

b) S.N.I que rota con igual ω que el riel

$$m \ddot{a}' = m \ddot{a} + 2m \dot{r}' \times \omega + m r' \times \dot{\omega} + m r' \perp \omega^2$$

$$m \ddot{a}' = F_v \hat{g}' - 2m \dot{x}' \omega \hat{g}' + m \omega^2 x' \hat{x}'$$

$$m \ddot{x}' \hat{x}' = F_v \hat{g}' - 2m \dot{x}' \omega \hat{g}' + m \omega^2 x' \hat{x}'$$

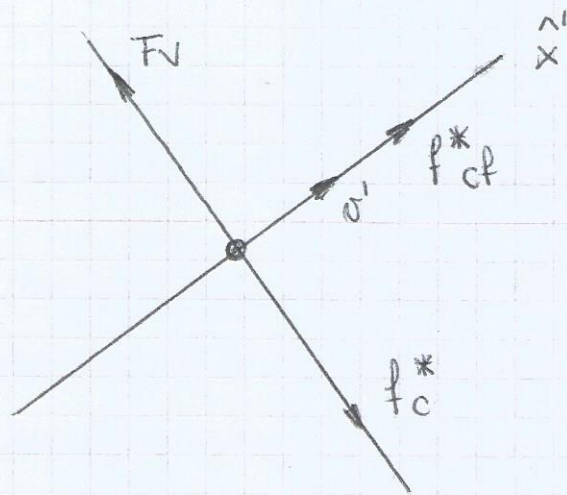
$$\ddot{x}' = m \ddot{x}' = m \omega^2 x'$$

$$\ddot{x}' - \omega^2 x' = 0$$

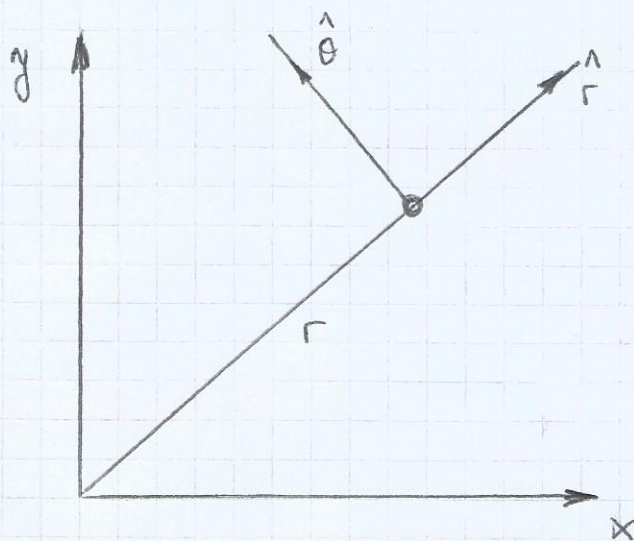
$$x' = A \cosh(\omega t + \varphi)$$

$$\hat{g}' = 0 = F_v - 2m \dot{x}' \omega$$

$$F_v = +2m \omega \dot{x}'$$



Desde el sistema inercial, uso coordenadas polares y considero
 $\hat{r} \equiv \hat{x}'$ y $\hat{\theta} \equiv \hat{y}'$



$$m\omega = F_v \hat{\theta}$$

$$m(\ddot{r} - r\dot{\theta}^2)\hat{r} + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} = F_v \hat{\theta}$$

$$\hat{r}) m(\ddot{r} - r\omega^2) = 0$$

$$\hat{r}) r(t) = A \cosh(\omega t + \varphi)$$

$$\hat{\theta}) m(2\dot{r}\omega) = F_v$$

$$\hat{\theta}) F_v = 2m\dot{r}\omega$$

$$\ddot{\mathbf{x}}' \cdot \hat{\mathbf{x}}' = \ddot{r} - 2\dot{x}'\omega\hat{y}' + \omega^2 x'x'$$

$$= (\ddot{r} - r\omega^2)\hat{r} + (2\dot{r}\omega)\hat{\theta} - 2\dot{x}'\omega\hat{y}' + \omega^2 x'x'$$

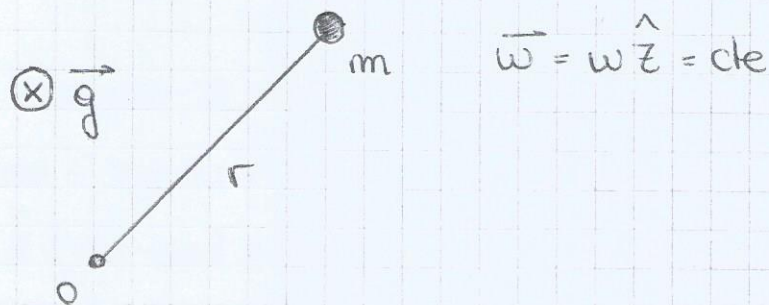
aceleración en polares

$$= (\ddot{x}' - x'\omega^2)\hat{x}' + (2\dot{x}'\omega)\hat{y}' - 2\dot{x}'\omega\hat{y}' + \omega^2 x'x'$$

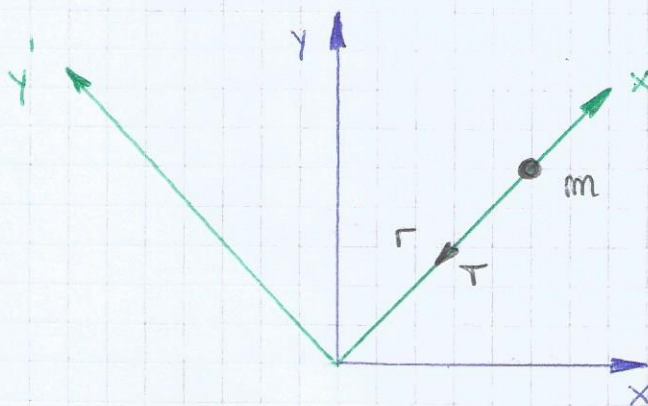
Centrípeta

Tangencial Coriolis Centrífuga

Ejemplo: Dado el siguiente sistema.



a) Describir el movimiento desde un S.N.I. que rota con ω .

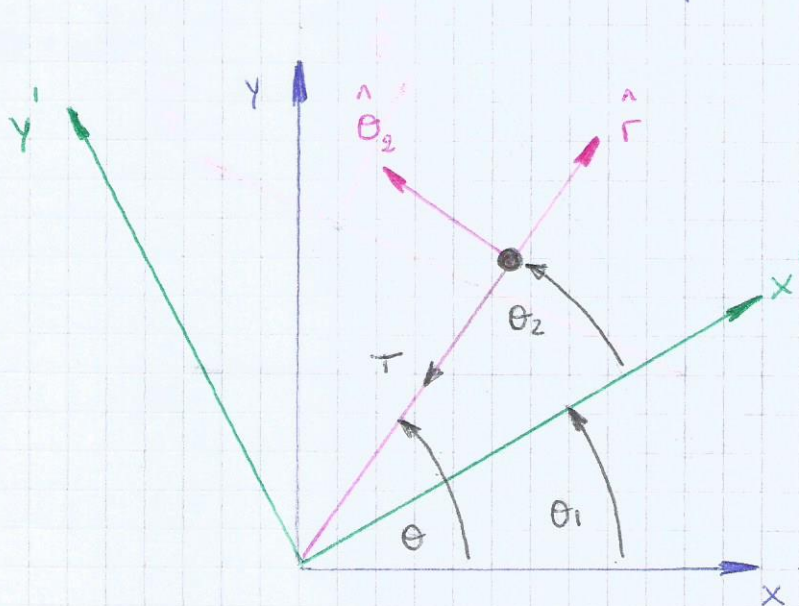


$$m\vec{a}' = m\vec{a} + m\vec{r} \times \dot{\vec{\omega}} + 2m\vec{v}' \times \vec{\omega} + m\vec{r} \perp \omega^2$$

$$0 = T + mR\omega^2 \hat{x}'$$

$$T = -mR\omega^2 \hat{x}'$$

b) Describir el movimiento desde un S.N.I. que rota con $\omega_1 < \omega$



$$\theta = \theta_1 + \theta_2$$

$$\theta_2 = \theta - \theta_1$$

$$\dot{\theta} = \omega$$

$$\dot{\theta}_1 = \omega_1$$

$$m \mathbf{a}' = m \mathbf{a} + \mathbf{f}_c^* + \mathbf{f}_{cf}^*$$

$$m \mathbf{a} = T = -m R \omega_1^2 \hat{r}$$

$$\mathbf{f}_c^* = m r \omega_1^2 \hat{r}$$

$$\left\{ \begin{array}{l} \mathbf{v}' = R \dot{\theta}_2 \hat{\theta}_2 \\ \mathbf{f}_c^* = 2 m \mathbf{v}' \times \omega_1 \end{array} \right.$$

$$\mathbf{f}_c^* = 2 m \mathbf{v}' \times \omega_1$$

$$\mathbf{f}_c^* = 2 m \dot{\theta}_2 R \hat{\theta}_2 \times \omega_1 \hat{z}$$

$$\mathbf{f}_c^* = 2 m \omega_2 \omega_1 R \hat{r}$$

$$\mathbf{v}' = \mathbf{v} - \omega_1 \times \mathbf{r}$$

$$\mathbf{v}' = \omega \times \mathbf{r} - \omega_1 \times \mathbf{r}$$

$$\mathbf{v}' = (\omega - \omega_1) \hat{z} \times R \hat{r}$$

$$\mathbf{v}' = (\omega - \omega_1) R \hat{\theta}_2$$

$$\mathbf{v}' = \dot{\theta}_2 R \hat{\theta}_2$$

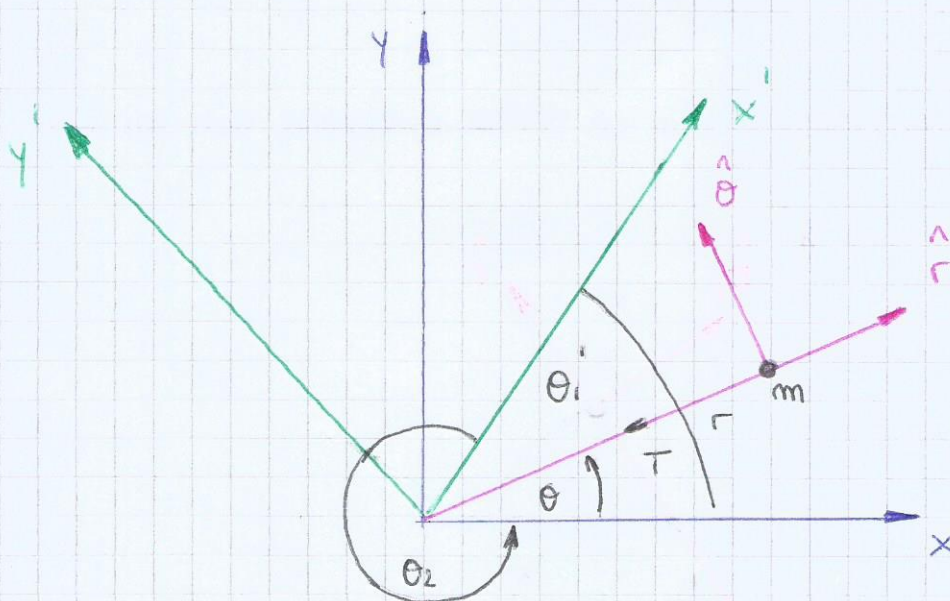
$$\mathbf{a}' = \frac{\mathbf{I}}{m} + 2(\omega - \omega_1) \omega_1 R \hat{r} + R \omega_1^2 \hat{r}$$

$$\mathbf{a}' = -\omega^2 R \hat{r} + 2(\omega - \omega_1) \omega_1 R \hat{r} + R \omega_1^2 \hat{r}$$

$$\mathbf{a}' = -R [\omega^2 - 2\omega \omega_1 + 2\omega_1^2 - \omega_1^2] \hat{r}$$

$$\mathbf{a}' = -R (\omega - \omega_1)^2 \hat{r}$$

c) Describir el movimiento desde un SNI que rota con $\omega_1' > \omega$



$$\theta_2 = 2\pi - (\theta_1 - \theta)$$

$$\dot{\theta}_2 = \dot{\theta} - \dot{\theta}_1' < 0$$

$$f_c^* = 2m \vec{v} \times \vec{\omega}$$

$$f_c^* = 2m (\vec{\omega} - \vec{\omega}') R \hat{\theta}_2 \times \omega' \hat{z}$$

$$f_c^* = 2m (\vec{\omega} - \vec{\omega}') R \omega' \hat{r}$$

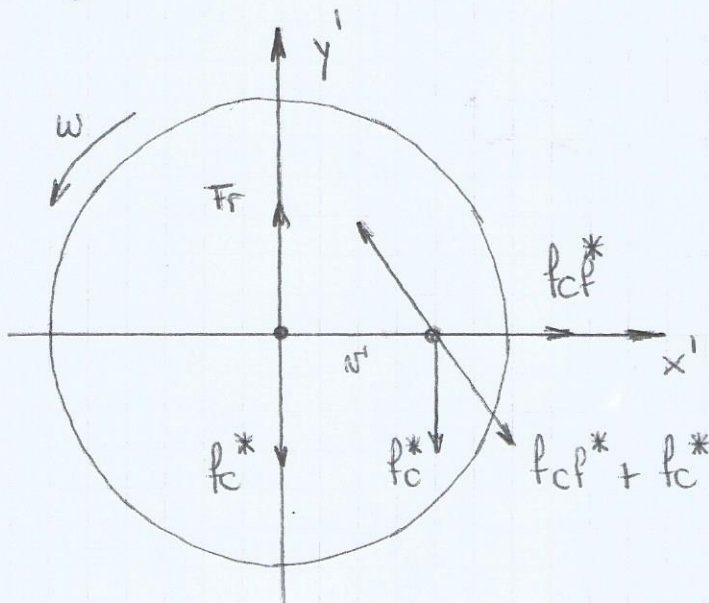
$$\vec{v}' = \vec{v} - \vec{\omega}' \times \vec{r}$$

$$\vec{v}' = \vec{\omega} \times \vec{r} - \vec{\omega}' \times \vec{r}$$

$$\vec{v}' = (\vec{\omega} - \vec{\omega}') \times \vec{r}$$

$$\vec{v}' = (\vec{\omega} - \vec{\omega}') R \hat{\theta}$$

Ejemplo:



$$\text{Si } r \perp \vec{\omega} \Rightarrow f_{cf}^* = 0$$

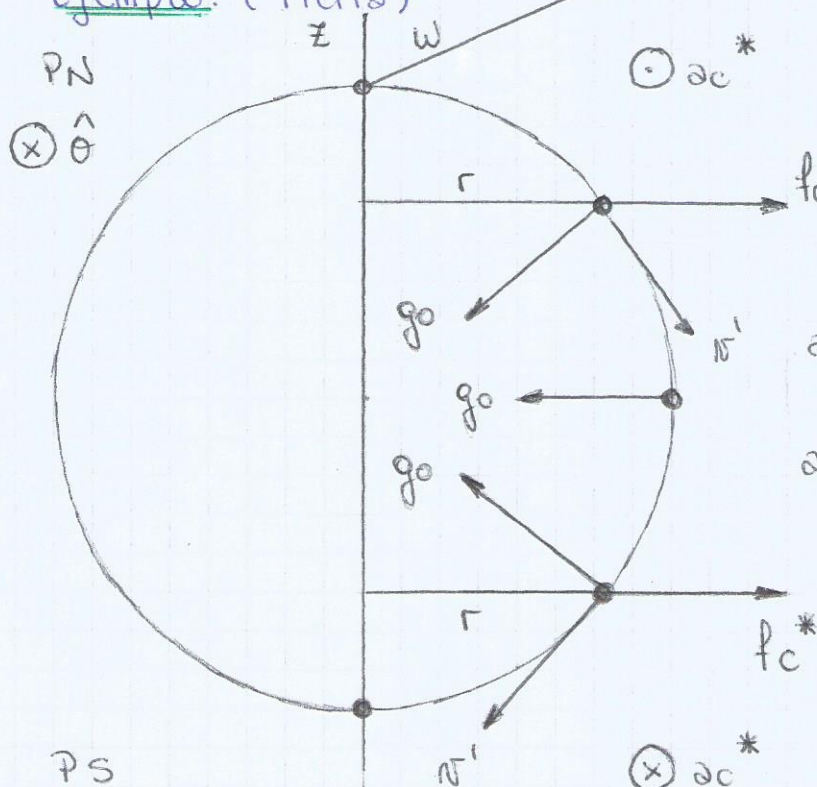
$$f_c^* = 2m \vec{v}' \times \vec{\omega}$$

$$f_c^* = 2m \dot{x}' \hat{x}' \times \omega \hat{z}$$

$$f_c^* = -2m \dot{x}' \omega \hat{y}'$$

$$\begin{cases} a_c^* = r \dot{\omega} \\ a_{cf}^* = 0 \end{cases}$$

Ejemplo: (Tierra)



$$\vec{a}' = g_0 + 2\vec{v}' \times \vec{\omega} + \vec{r} \omega^2$$

$$a_c^* = 0$$

$$a_{cf}^* = r \dot{\omega}$$

$$\vec{v}' = (\vec{v}_r' \hat{r}' - \vec{v}_z' \hat{z}')$$

$$\vec{v}' = (-\vec{v}_r' \hat{r}' - \vec{v}_z' \hat{z}')$$

Ejemplo: (Proyección)

